

Module 4 - Probability and Statistics

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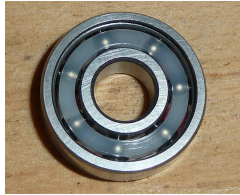
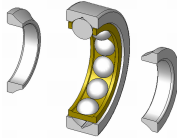
- Topic 1 - Histograms and Probability Density Functions
- Topic 2 - Characterizing a Population of Data
- Topic 3 - Z Table Examples

Topic 1 - Histograms and Probability Density Functions

- A Population of Data
- Randomly Distributed Data
- Frequency Histogram
- Probability Density Function

A Population of Data

Consider a manufacturer that makes 100000 of ball bearings in a set.



- How does the manufacturer ensure the quality of the product?
- How the does the seller communicate this to the buyer?
- How does the engineer use this information?

Images: [Wikipedia](#)

A Population of Data

The data set, or population, is generated by taking measurements of individuals chosen randomly from the entire population.

Sampling refers to repeated measurements of the **measured variable** under fixed operating condition.

$$X = \{x_1, x_2, x_3, \dots, x_i, \dots, x_{n-1}, x_n\}$$

Randomly Distributed Data

Our discussion will assume that the values in the data set are **randomly distributed** about a mean value. It is important to consider what this means.

Frequency Histogram

"A **histogram** is an accurate representation of the distribution of numerical data. It is an estimate of the probability distribution of a continuous variable (CORAL) and was first introduced by Karl Pearson.[1] It differs from a bar graph, in the sense that a bar graph relates two variables, but a histogram relates only one." - Wikipedia